

SUYOG HANMANT BARGULE

Artificial Intelligence and Computer Vision Engineer

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 <https://github.com/suyogbargule>

 <https://suyogbargule.github.io/portfolio>

PROFILE SUMMARY

- Innovative and results-driven Generative AI and Computer Vision Engineer** with over **3+ years** of hands-on experience in **designing, developing, and deploying** cutting-edge **AI solutions**. Proven expertise in **face recognition, neural networks, image processing, generative AI, and real-time model optimization** consistently delivering measurable improvements in **system accuracy** and **operational efficiency**.
- Demonstrated success in building scalable and intelligent systems across various domains, including **surveillance, traffic monitoring, and assistive technologies**. Adept at creating smart **AI agents** and **conversational chatbots** that enhance user experience and drive **automation**. Passionate about transforming complex challenges into impactful, **real-world solutions** through **AI innovation**.

WORK EXPERIENCE

Computer Vision Engineer
Futops Technologies Pune  Oct 2022-current  Pune Maharashtra, India

- Developed a company-specific **chatbot using Lingcine and DeepSeek** models, deployed locally with Ollama. Leveraged **CUDA** for real-time inference on a **14B parameter model**. Integrated the chatbot exclusively with the company website to enable seamless interaction with end users.
- Developed and deployed a traffic surveillance system. Detection accuracy for **no-helmet, triple riding, and red-light violations** increased by **81%**. Implemented a **best snapshot** technique for object detection across multiple frames, **raising accuracy from 40% to 75%** and received a **SPOT award** for this achievement.
- Created an AWS-based solution that increased the **independence of blind users by 80%**, utilizing real-time camera and headphone integration. The solution **achieved 82% accuracy** in detection and guidance for blind individuals.
- Enhanced a face recognition surveillance system with real-time detection, achieving **90% accuracy in face detection** and **95% accuracy in face recognition**. The system includes age, gender, mask detection, and feature extraction capabilities, with per-face processing times take to **30-35 milliseconds**.
- Implemented panoramic imaging for **360° CCTV cameras, increasing coverage by 100%** and generating **night-time panoramas** in low-light conditions using **OpenCV and GStreamer** and received a **SPOT award** for this achievement.

AI Research
AI4SEE.pvt.ltd  Feb 2021-Sep 2022  Bangalore Karnataka, India

- Rope Jump Counting App:** Developed a mobile application using Flutter that counts rope jumps, leveraging a TensorFlow Lite model for real-time motion detection and analysis.
- Farmer Product Sales App:** Developed a mobile app that enabled farmers to sell products directly to retail customers, increasing sales by 20%. Implemented a regression model to forecast product prices five days in advance, improving pricing accuracy and sales efficiency.

SKILLS

- programming languages :-** Python, C++ .
- Technologies :-** Linux, Docker, ZeroMQ, AWS Rekognition, SageMaker, S3, EC2, Azure AI, Azure. PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, ONNX, OpenCV, OpenVINO, CUDA, NumPy, Pandas. CNN, GAN, NLP, LLM(Large Language Models), RAG(Retrieval-Augmented Generation), LangChain.
- Specialties :-** Deep Learning, Machine Learning, Generative AI, Vision Transformers. Object Classification and Detection, Image Processing, Image Segmentation. Time Series Analysis, Natural Language Processing (NLP), LLaMA, deepseek model. Generative AI Applications (e.g., text-to-image models, GPT-based solutions). Data Manipulation and Analysis
- Soft skills :-** Teamwork and Collaboration,Adaptability,Analytical Thinking,Problem-Solving,Effective Communication
- Core competencies :-** Agile methodologies, Performance Optimization, Containerization .

LANGUAGES

English, Hindi, Marathi

EDUCATION

Master of Computer Application
Shivaji University Computer Science Department | Graduation Year: 2019

CERTIFICATIONS

Data Science Jan 2019-May 2019
ExcelR Solution Bangalore.